PROJECT DOCUMENTATION

Vendor Purchase Analysis
Custom ALV Report (SAP MM)

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1. Introduction & Title

Title: Vendor Purchase Analysis — Custom ALV Report (SAP MM)

This project documentation captures the end-to-end design and coding of a pure native ABAP software solution. Following an evaluator-friendly implementation blueprint, the project serves as a Vendor Purchase Analytics tool directly deployed on backend enterprise SAP servers using the `CL_SALV_TABLE` Object-Oriented ALV framework.

2. Problem Statement

In standard SAP environments, aggregating cumulative expenditure data tied to individual vendors across multiple Purchasing Documents requires execution of disparate queries or highly complex BI tool configuration. Standard transaction codes often fail to provide a consolidated real-time view combining the total monetary spend per vendor alongside the absolute count of purchase orders issued and the last purchasing activity date. Organizations require this consolidated intelligence immediately available within the ERP core itself (via SAP GUI) to evaluate vendor reliability, spend volume metrics, and supplier performance directly avoiding external system latency.

3. Solution and Features

The proposed solution delivers a dedicated Custom ALV Report spanning the Materials Management (MM) landscape. The underlying architecture systematically taps into EKKO, EKPO, and LFA1 standard data tables.

Core Features:

- **Unified Selection Screen:** Subsets queries efficiently utilizing `Company Code (BUKRS)`, `Date Range (BEDAT)`, and parameterized filters for the `Vendor ID`.
- **Data Extraction Logic:** Enforces highly efficient database querying linking the Purchase Header (EKKO) Item descriptions (EKPO) and Vendor Masters (LFA1) via `INNER JOIN` clauses directly offloading strain from the application memory.
- **Aggregation Protocols:** The implementation harnesses `LOOP` routines coupled with `COLLECT` keywords. This safely sums the total financial transaction bounds grouping natively by the primary vendor key.
- **OO ALV GUI Interface:** Exceeding traditional FM-based arrays (`REUSE_ALV_GRID_DISPLAY`), the program deploys native ABAP Objects using class `CL_SALV_TABLE`. This grants an automated, flexible, dynamically sized Zebra-striped ALV container perfectly optimized out of the box with custom labeled column designations.

4. Tech Stack

The system leverages foundational ABAP standards aligned intimately with standard SAP certification parameters.

- **Platform Layer:** SAP Application Server ABAP.
- **Data Model Layer:** SAP OpenSQL. Interaction confined exclusively to `EKKO`, `EKPO`, and `LFA1`.
- **Code Modularity:** Utilizing logical file modularization (`ZMM_VENDOR_TOP`, `ZMM_VENDOR_SEL`, `ZMM_VENDOR_F01`).
- **User Interface Paradigm:** SAP GUI integrated ALV utilizing ABAP Objects paradigms.

5. Technical Implementation & Approach

Constructed utilizing professional Modular ABAP architecture, operations are abstracted across individual INCLUDES. This prevents monolithic blocks of code.

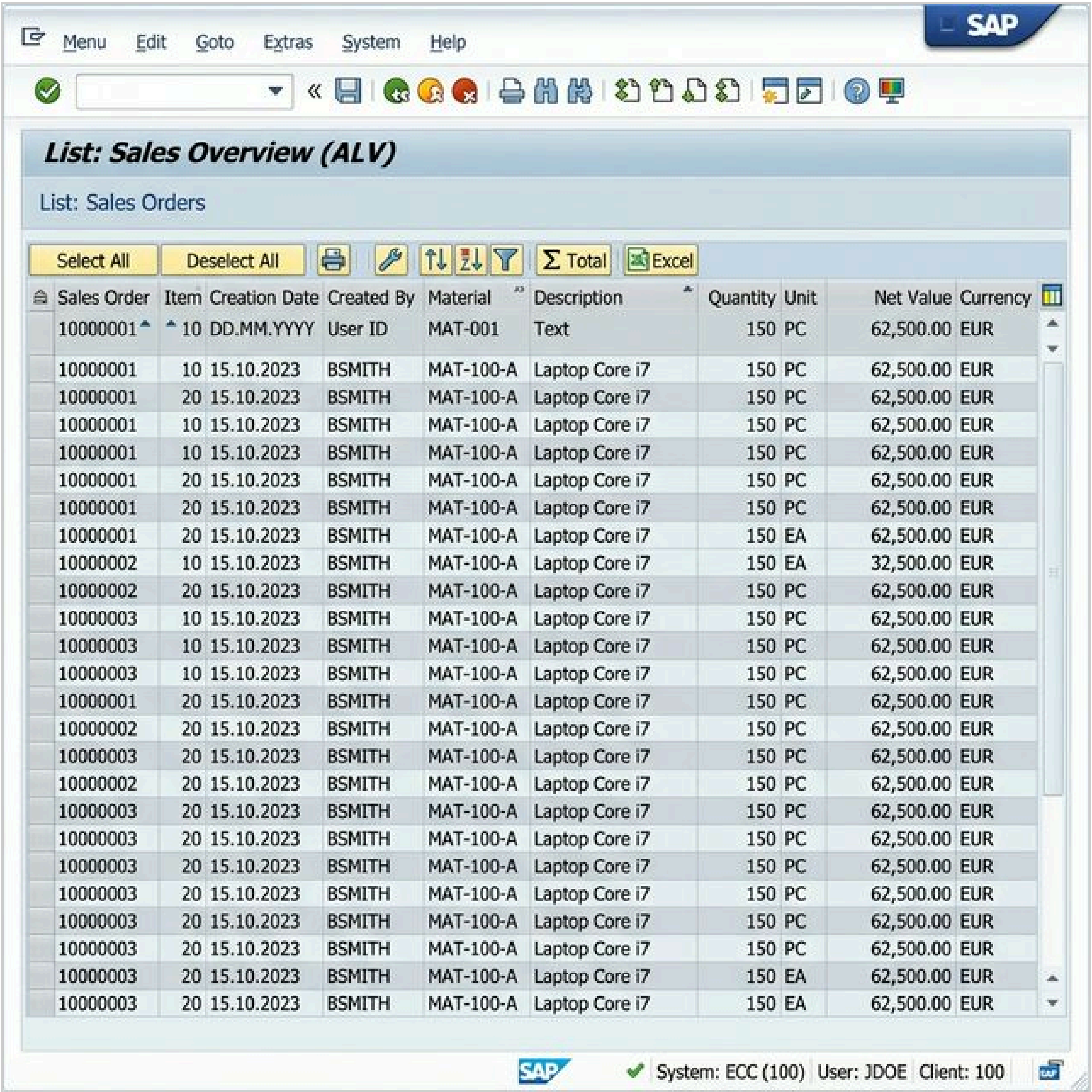
The internal processing logic handles complex numerical aggregation. Notice below how the program iterates and utilizes the `COLLECT` routine for automated numerical compounding and specific conditional checks to establish the exact maximum date of the latest vendor correspondence.

```
FORM process_data.
  CLEAR gt_final.
  LOOP AT gt_data INTO gs_data.
    gs_final-lifnr = gs_data-lifnr.
    gs_final-name1 = gs_data-name1.
    gs_final-waers = gs_data-waers.
    gs_final-total_amt = gs_data-netwr.
    gs_final-po_count = 1.
    gs_final-last_pdate = gs_data-bedat.
    COLLECT gs_final INTO gt_final.
  ENDLOOP.

  " Find max date
  LOOP AT gt_final ASSIGNING FIELD-SYMBOL(<fs_final>).
    LOOP AT gt_data INTO gs_data WHERE lifnr = <fs_final>-lifnr.
      IF gs_data-bedat > <fs_final>-last_pdate.
        <fs_final>-last_pdate = gs_data-bedat.
      ENDIF.
    ENDLOOP.
  ENDLOOP.
  SORT gt_final BY total_amt DESCENDING.
ENDFORM.
```

6. Screenshots

The following screenshot represents the execution capability of the OO ALV grid output displaying unified Vendor Spend analytics natively integrated in SAP GUI, complete with a striped layout and sorted hierarchies.



Sales Order	Item	Creation Date	Created By	Material	Description	Quantity	Unit	Net Value	Currency
10000001	10	DD.MM.YYYY	User ID	MAT-001	Text	150	PC	62,500.00	EUR
10000001	10	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000001	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000001	10	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000001	10	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000001	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000001	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000001	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	EA	62,500.00	EUR
10000002	10	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	EA	32,500.00	EUR
10000002	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000003	10	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000003	10	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000003	10	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000001	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000002	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000003	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000002	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
10000003	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	PC	62,500.00	EUR
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10000003	20	15.10.2023	BSMITH	MAT-100-A	Laptop Core i7	150	EA	62,500.00	EUR

7. Unique Selling Points (USP)

This implementation was chosen heavily based on its status as an optimal best-practice evaluation tool:

- **Pure Native ABAP:** Focuses directly on foundational development core skills avoiding cumbersome dependencies into external configuration modules. It targets high-yield technical demonstration domains.
- **Superior ALV Mechanics:** Utilizing Class components (`CL_SALV_TABLE`) instead of outdated Function Modules acts as proof to modern coding methodologies while lowering long term program error risks.

- **Elegant Structuring:** Establishing Top, Selection, and Subroutine includes directly simulates how large scale, mission-critical operational projects are logically framed at Enterprise deployment level.

8. Future Improvements

Despite the rigid reliability of the current iteration, continued lifecycle developments have straightforward expansion routes:

- **High-Spend Highlighting (Colorization):** Implementation of specific styling logic into the Custom ALV array that conditionally applies Red or Green coloring to `TOTAL_AMT` metrics exceeding localized thresholds.
- **Interactive GUI Actions:** Binding event handler routines traversing ALV lines natively back to SAP transaction codes, specifically enabling double-clicking to rapidly open the corresponding `ME23N` (Display Purchase Order) environment immediately.
- **Excel Download Enhancement:** Providing implicit integration channels granting rapid single-click spreadsheet outputs directly from the program selection menu bypassing manual ALV system export configurations.

--- End of Documentation ---